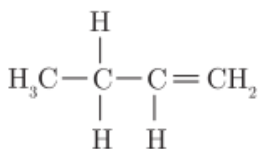
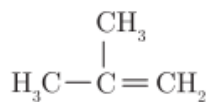
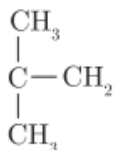
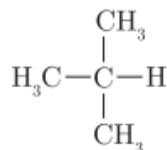


Chapter

Organic Chemistry

1. The structures of four hydrocarbons are shown below:



How many isomers of butene are there?

SQP 2025

- (a) 1
- (b) 2
- (c) 3
- (d) 4

2. Unsaturated hydrocarbons undergo: MAIN 2024

- (a) Addition reaction
- (b) Substitution reaction
- (c) Oxidation reaction
- (d) Redox reaction

3. Reaction of an alcohol with a carboxylic acid in the presence of concentrated H_2SO_4 is termed as: MAIN 2024

- (a) Halogenation
- (b) Esterification
- (c) Hydrogenation
- (d) Dehydrohalogenation

4. Conversion of Ethanol to Ethene by the action of concentrated sulphuric acid involves: MAIN 2024

- (a) Dehydration
- (b) Dehydrogenation
- (c) Dehydrohalogenation
- (d) Hydrolysis

5. The reaction of alcohol with acetic acid in the presence of concentrated sulphuric acid is known as MAIN 2021

- (a) Distillation
- (b) Esterification
- (c) Saponification
- (d) Condensation

6. Methyl alcohol reacts with in the presence of conc. H_2SO_4 to give an ester. COMP 2023

- (a) Phenol
- (b) Benzoic acid
- (c) Ethanoic acid
- (d) Benzene

7. Which statement about ethanoic acid is correct ? COMP 2023

- (a) It reacts with ethanol to form a sweet smelling Compound.
- (b) It contains three carbon atom per molecule.
- (c) It is insoluble in water.
- (d) It contains five hydrogen atoms per molecule.

8. Molecular formula of butane is MAIN 2022

- (a) C_4H_{10}
- (b) C_5H_{10}
- (c) C_4H_8
- (d) C_4H_{12}

9. Which of the following Compound has highest boiling point ? SQP 2023

- (a) 2-methylbutane
- (b) n-pentane
- (c) n-hexane
- (d) 2,2-dimethyl propane

10. When acetic acid is heated with phosphorus pentaoxide is formed. SQP 2022

- (a) acetic anhydride
- (b) aliphatic Compound
- (c) cellulose acetate
- (d) methanoic acid

11. Alkynes undergo ____ [addition/ substitution] reactions. SQP 2025

12. State two Main groups of aliphatic hydrocarbons with examples. MAIN 2022

13. Draw the structural formulae of the two isomers of butane. Give the correct IUPAC name of each isomer. COMP 2023

14. Choose the organic compound from the list given below to answer the following question: Ethene Ethanoic acid Ethanol Methanal MAIN 2024

- (a) The compound which does not have double bond in its structure.
- (b) The compound which in its pure form turns into an ice like solid on cooling.
- (c) The compound which is used for artificial ripening of fruits.

15. Draw the structural diagram for the following organic compounds:

- 1. 3-methyl pentane
 - 2. propyne
 - 3. Methanal
- MAIN 2024**

16. What is the Main difference in the structural formula of the Compounds n-pentane, iso-pentane and neo-pentane ? MAIN 2021

17. State the terms for the following: MAIN 2024

- (a) The group obtained by removing one hydrogen atom from the parent alkane.
- (b) Two metal plates or wires through which the current enters and leaves the electrolytic cell.
- (c) The amount of substance which contains the same number of units as the number of atoms in carbon-12.
- (d) The tendency of an atom to pull a shared pair of electrons towards itself in a compound.
- (e) The formula which represents the simplest ratio between the atoms of elements present in a compound.

18. What are the final products when acetylene (ethyne) adds to MAIN 2021

- (i) Iodine
- (ii) Chlorine
- (iii) Bromine (liquid)
- (iv) Hydrogen ?

Solution

1. Thus (b) is correct option.

2. (a) Addition reaction

Unsaturated hydrocarbons like alkenes and alkynes contain double or triple bonds which are highly reactive. They undergo addition reactions, where the unsaturated bond is broken and new atoms or groups are added, forming a saturated molecule.

3. (b) Esterification

When an alcohol reacts with a carboxylic acid in the presence of concentrated sulphuric acid (H_2SO_4), an ester is formed along with water. This reaction is known as esterification.

4. (a) Dehydration

When ethanol is heated with concentrated sulphuric acid, a molecule of water is eliminated from ethanol to form ethene. This process is called dehydration, as it involves the removal of water from the alcohol.

5. (b) Esterification

When an alcohol reacts with acetic acid in the presence of concentrated sulphuric acid (H_2SO_4), an ester is formed along with water. This reaction is called esterification.

6. (c) Ethanoic acid

Methyl alcohol (CH_3OH) reacts with ethanoic acid (CH_3COOH) in the presence of concentrated sulphuric acid (H_2SO_4) to form methyl acetate ($\text{CH}_3\text{COOCH}_3$) and water. This is an example of esterification.

7. (a) It reacts with ethanol to form a sweet smelling compound

Explanation:

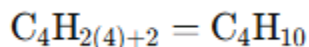
Ethanoic acid (CH_3COOH) reacts with ethanol ($\text{C}_2\text{H}_5\text{OH}$) in the presence

of concentrated sulphuric acid to form ethyl acetate ($\text{CH}_3\text{COOC}_2\text{H}_5$), which is a sweet-smelling ester, along with water.

8. (a) C_4H_{10}

Explanation:

Butane is an alkane with four carbon atoms in a straight chain. Being an alkane, it follows the general formula $\text{C}_n\text{H}_{2n+2}$, so for $n = 4$:



9. (c) n-Hexane

Boiling point depends on the surface area and intermolecular forces. Straight-chain alkanes have a larger surface area and stronger van der Waals forces than branched alkanes, resulting in a higher boiling point. Among the given compounds, n-hexane is a straight-chain alkane and thus has the highest boiling point.

10. (a) Acetic anhydride

When Acetic acid is heated in the presence of Phosphorous pentoxide, it results in the removal of water and forms Acetic anhydride.

11. addition

12. Aliphatic hydrocarbons are mainly classified into two groups:

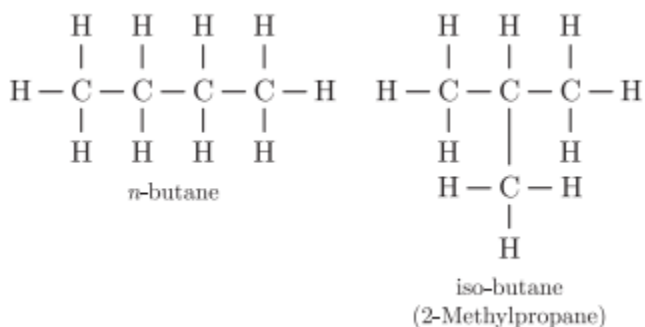
1. **Saturated aliphatic hydrocarbons (Alkanes):** These contain single bonds only.

Examples: Methane (CH_4), Ethane (C_2H_6)

2. **Unsaturated aliphatic hydrocarbons:** These contain double or triple bonds.

Examples: Ethene (C_2H_4), Ethyne (C_2H_2)

13.



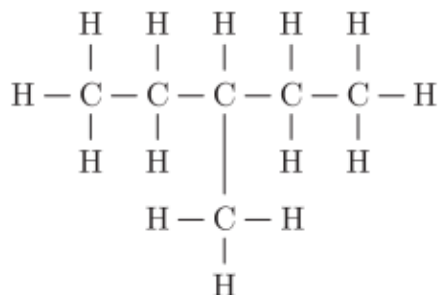
14. (a) The compound which does **not** have a double bond: Ethanol ($\text{C}_2\text{H}_5\text{OH}$)

(b) The compound which **turns** into an ice-like solid on cooling: Ethanoic acid (CH_3COOH)

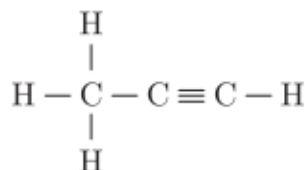
(c) The compound which is **used** for artificial ripening of fruits: Ethene (C_2H_4)

15.

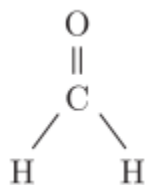
1. 3-methylpentane



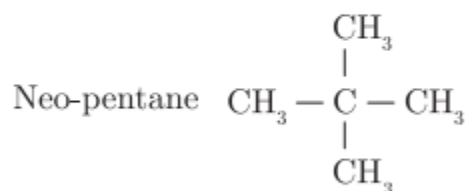
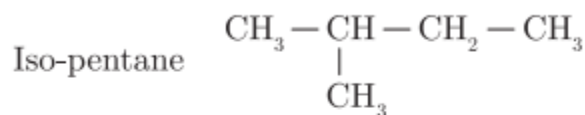
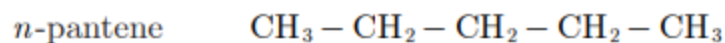
2. Propyne



3. Methanal



16.



The difference in the three isomers is that they have different lengths of chain of C-atoms.

17. (a) Alkyl group

(b) Electrodes (cathode and anode)

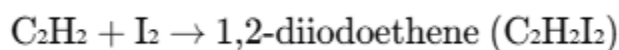
(c) Mole

(d) Electronegativity

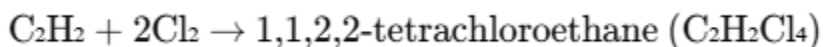
(e) Empirical formula

18. When acetylene (C_2H_2) undergoes addition reactions, the final products are:

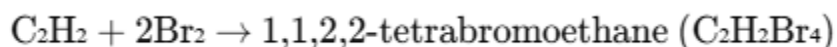
(i) With iodine (I_2):



(ii) With chlorine (Cl_2):



(iii) With bromine (Br_2):



(iv) With hydrogen (H_2):

